

FILTER DESIGN TEST CASE

High-Pass Filter - Band 26 (850 MHz) for WiFi 6E Coexistence Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P4 - Low

1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 26 (850 MHz) for WiFi 6E Coexistence Filter. The filter is designed on ScAlN on Si substrate with a target center frequency of 3973.2 MHz and bandwidth of 47.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 26 (850 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: ScAlN on Si
- Package: WLP (Wafer Level Package)
- Design Tool: Sonnet EM Suite v16
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the High-Pass Filter design targeting Band 26 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3973.2 MHz
Bandwidth (3 dB):	47.1 MHz
Insertion Loss Target:	≤ 1.1 dB
Return Loss Target:	≥ 18.0 dB
Out-of-Band Rejection:	≥ 50.0 dB
Isolation (Duplexer):	≥ 43.0 dB
Q Factor:	6998
Temperature Coefficient:	-12.4 ppm/°C
Die Size:	1.1 x 0.9 mm
Wafer Size:	8-inch
Number of Resonators:	9
Electrode Material:	Mo

Passivation: SiO2

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the High-Pass Filter template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3973.2 MHz, BW 47.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, ScAlN on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11920 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.1 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 71 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 6.2 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-05-18
Devices Tested: 25
Insertion Loss (meas): 0.85 dB
Return Loss (meas): 20.3 dB
Rejection (meas): 53.2 dB
Isolation (meas): 47.7 dB
Q Factor (meas): 6998
Group Delay Variation: 7.0 ns
Power Handling: +35 dBm
Die Yield: 87.0%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The High-Pass Filter design demonstrated below-target performance for Band 26 (850 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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